

Optical Transparency of a Heterogeneous Pan-European Network

Pierre Péloso, Denis Penninckx, Magali Prunaire, and Ludovic Noirie

Abstract—We have experimentally shown that the physical design of a transparent network should be tailored to both the network topology and the network traffic matrix. For a pan-European network for which 85% of the traffic was shorter than 2000 km, almost all connections may be transparent; with very simple engineering rules (erbium-doped fiber amplifiers only, no power equalization except for the per-band equalization in the node, constant static dispersion management whatever the ingress and egress nodes, and whatever the wavelength. . .). We have also experimentally shown how dispersion management should take into account unavoidable inaccuracies that are to be found in the field.

Index Terms—Optical hybrid networks, transparency, waveband, wide area networks (WANs).

I. INTRODUCTION

IN today's optical networks, the signal is regenerated at each node and the network is said to be opaque. Thus, the optical connections are point-to-point. Removing most optoelectronic regenerators used in optical networks by switching directly in the optical domain would strongly reduce hardware costs. However, this concept, called transparency, requires the capability to cascade at least a few links.

One way to do so is to insert specific devices, such as Raman amplification, dynamic gain equalizers [1], or dynamic dispersion compensation, to increase propagation distances and cope with a group of historically different signals. But all these devices have a cost competing with the savings of optoelectronic interfaces. Alternatively, it was suggested to use a very simple design such as simple dispersion-management rules [2], [3]; no-dynamic-gain equalization [2]; or erbium-doped fiber amplifiers. The remaining traffic could be conveyed with little remaining optoelectronic regeneration capabilities. (This concept is called hybrid transparency, in between transparent and opaque.)

Although some took interest in homogenous networks [4], heterogeneous ones were little considered. Where amplifier spans are not unique over the whole network (except for [5] and [6]), none systematically explored all the connections to quantify the ratio of connections that can be established transparently. To address these issues, we have *experimentally* simulated every connection in a hybrid network, featuring most European capitals.

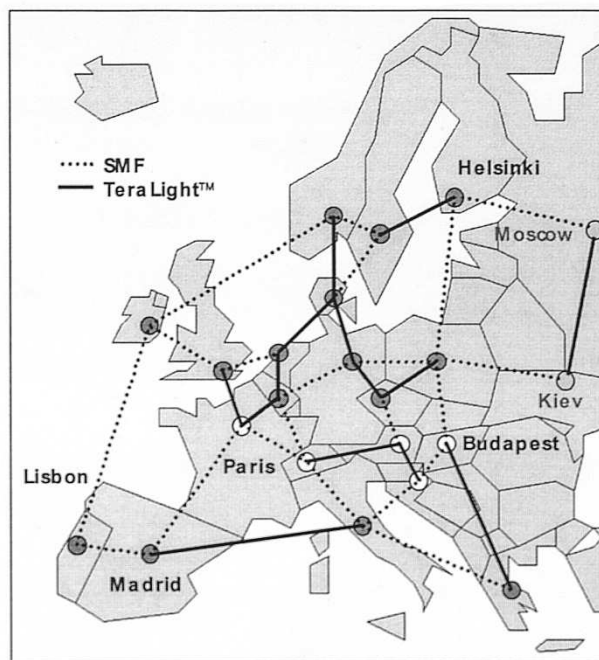


Fig. 1. Pan-European network topology (extended with Kiev and Moscow).

This paper is focused on the physical design of the network. We will not consider the routing and wavelength assignment algorithms, although a realistic network topology with a realistic traffic matrix is used. The traffic matrix is normalized: what matters is the ratio of the traffic for any given connection. The absolute value is of no importance for the physical design: if one fiber is not sufficient between any two nodes, an additional fiber is considered to be used. Nevertheless, the physical design of the network should be adapted to the connection lengths. Hence, to approximately determine them, we have considered a shortest kilometric path.

In the following, we will describe the heterogeneous backbone network and the engineering rules applied to its physical design. We will then explain how we used the *multiloop* [7] to test these rules and challenge their pertinence versus network transparency before concluding on the crossed influence of network (topology and traffic) and physical design dispersion rules on the ratio of transparently supported traffic.

II. PRESENTATION OF THE NETWORK USED HEREAFTER

A. Network Topology

A European meshed backbone network, which was already presented in [8], was taken into consideration (Fig. 1). It is not an existing network, but is a good representation of any

Manuscript received June 27, 2003; revised September 15, 2003.

The authors are with Alcatel Research and Innovation, Marcoussis Cedex 91461, France (e-mail: pierre.peloso@alcatel.fr; denis.penninckx@alcatel.fr).

Digital Object Identifier 10.1109/JLT.2003.822831

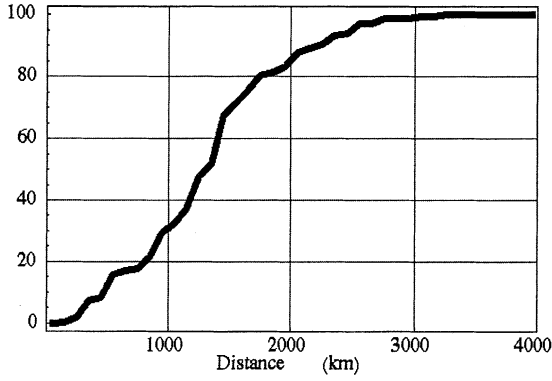


Fig. 2. Cumulated probability of connection distance weighted by the traffic.

pan-European network with typical node connectivity and link lengths. The 20 nodes are located at capitals of European countries with a meshed topology composed of 32 links. Every connection in this network is taken as the shortest kilometric path between ingress and egress. This network is named original network in the following in contrast with the extended network where two nodes (Kiev and Moscow) and three links were added. The extended network will be of use later in the paper.

The heterogeneity of this network has been arbitrarily chosen and comes from the difference in the amplifier spans composing the links. Either spans of 75 km of standard single mode fiber (SMF) with a chromatic dispersion of +17 ps/(nm.km) and an average channel launched power of +1 dBm, or spans of 100 km of TeraLight fiber at +8 ps/(nm.km) and an average power channel launched power of +5 dBm. We have randomly drawn the fiber type composing each link (Fig. 1). These links are then defined as a succession of a given number of spans of a given fiber type and a connection between any two nodes is a concatenation of links and nodes.

B. Traffic Matrix

The traffic matrix of this network is defined proportionally to the populations of the countries of both end nodes of every connection [8], out of which was plotted the cumulated probability of connection distance (Fig. 2). For example, 85% of the connections range between 0 and 2000 km.

Tightened engineering rules able us to reach 6000 km regardless of whether or not the wavelengths are not appropriated with such a traffic matrix, especially that it would induce an extra cost on the whole network. Some simpler ones, which allow approximately 2000 km, would be clearly sufficient. According to that observation, they were chosen as described hereafter. Moreover, in a point-to-point transmission, all the wavelengths have to bridge the distance while in a transparent network; some most favored wavelengths may be used for the longest connections while the others maybe used for shorter connections.

C. First Set of Engineering Rules

The set of engineering rules used in this paper is committed to use the simplest possible options in order to get low-cost solu-

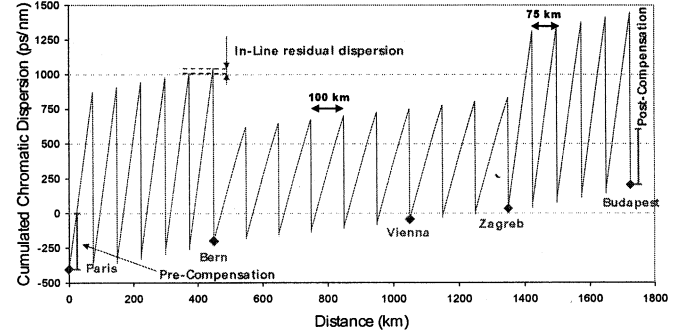


Fig. 3. Focus on the Paris to Budapest connection: dispersion map.

tions and an easy-to-manage system, whereas being consistent with the criteria proposed in [2] and [3]. The engineering rules applied to the network in a first step are the following.

- The network is hybrid, transparent at the waveband level [9] while allowing regeneration at the wavelength level (see node architecture, Fig. 4). This partial regeneration capability avoids constraining physical design rules.
- 10-Gb/s enhanced forward error-correction code (EFEC). Hence, the line bit rate is 10.709 Gb/s. $2 \cdot 10^{-3}$ bit error rate (BER) before EFEC is sufficient to get 10^{-3} after correction.
- Nonreturn-to-zero (NRZ) modulation format.
- Spectrum composed of eight wavebands of four wavelengths with 100-GHz spacing and a skipped wavelength in between wavebands, filling the whole C-Band (Fig. 5).
- EDFAs only.
- Launched powers (SMF line: +1 dBm/ch; TeraLight line: +5 dBm/ch; SMF compensation: -5 dBm/ch; TeraLight compensation: -2 dBm).
- Neither channel preemphasis at ingress nor spectral equalization in line, out of the per-waveband equalization, is achieved in the nodes.
- Constant chromatic dispersion-management rules are experimental simulations that were conducted with very simple chromatic dispersion-management rules constant precompensation and inline residual dispersion and postcompensation, regardless of the ingress and egress nodes and the wavelength. These rules avoid any dynamic adjustment of dispersion at the system level, making for a simple system. Nevertheless, to be efficient, the dispersion-management rules need to be tolerant to transmission distance and fiber type. Hence, the following value were chosen: precompensation (at ingress): -400 ps/nm; inline residual dispersion (cumulated dispersion of a span): +30 ps/nm/span (whatever the fiber type); postcompensation (at egress): +400 ps/nm [2].

These features are illustrated in a typical connection (Paris to Budapest, Fig. 1 white-colored nodes) with the dispersion map presented in Fig. 3.

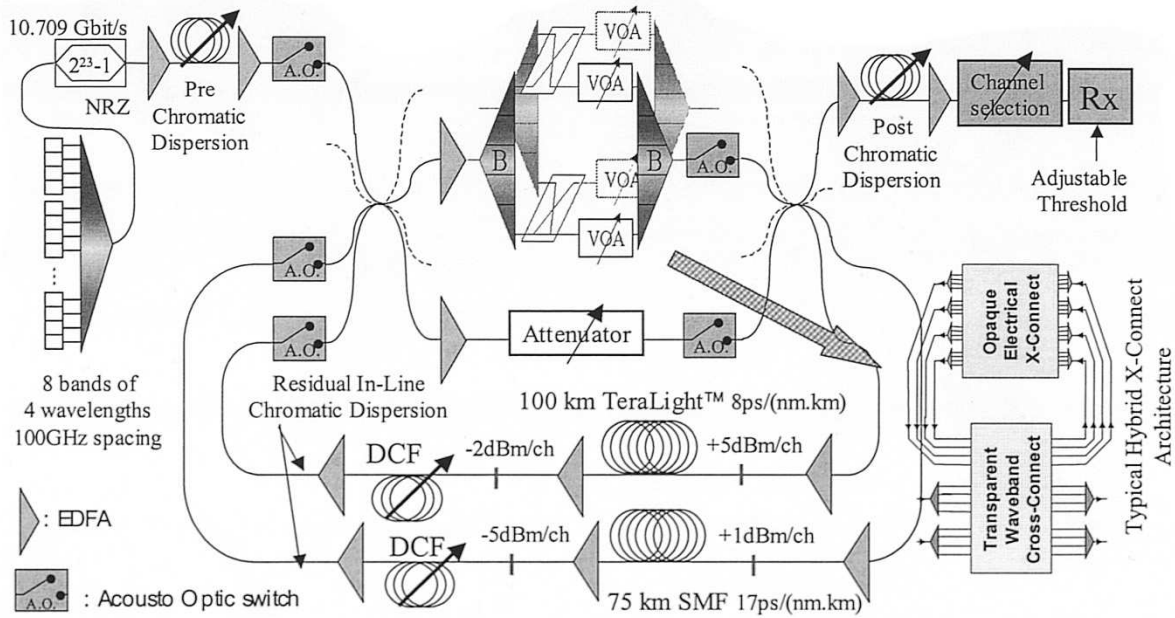


Fig. 4. Multiloop experimental setup.

III. EXPERIMENT

A. Experiment Principles

Quantifying the ratio of traffic that can be established transparently in a given network is what motivates the approach framing the work presented here.

To achieve such a goal, every connection of the network has to be reproduced in the laboratory and the BER of every wavelengths has to be measured. Under the hypothesis described above, the number of connections is 380 (20 nodes) and the number of wavelengths is 32, which makes more than 12 000 BER to measure. These experimental data analyzed with respect to the traffic matrix of the networks gives the ratio of transparently supported traffic in the considered network.

Hereafter is described how we managed to reproduce every connections of the above presented network in the laboratory.

B. Multiloop Principle and Configuration

The experimental tool—the multiloop—is a flexible recirculating loop [10], [11] with which we can select live, between different transmission lines, the one in which the signal travels every roundtrip, with the possibility to pass through an optical node in between roundtrips.

Two lines are equipped with both the amplifier spans described in Section II-C. Fast acousto-optical switches select the line in which the signal travels, depending on the index of the roundtrip.

In hybrid networks, the signal goes through transparent nodes. To simulate this feature, the multiloop has to give the opportunity to get (or not) through such a node in between roundtrips. This feature is realized by an extra pair of short lines added between the two ends of the main lines; only one contains a transparent node. It is then possible to simulate a connection such as the Paris to Budapest one: six roundtrips on the SMF line, passing through the node; six roundtrips on the TeraLight

line, passing through the node; then three TeraLight roundtrips, through the node one last time; and 5 SMF roundtrips.

In the like of a recirculating loop, the amplifiers need to be continuously saturated. As both lines are not the same length, the number of following bits (bit stream) that can be measured is function of the shorter line. Of course, this implies that, in the longer line, the bit stream is partially repeated to maintain the EDFA's saturation. This repetition of the bit stream happens when a changing of line occurs (toward the longer one). Actually, it is the shorter lines that spreads more than a roundtrip in the longer one.

C. Tuning Details of the Experimental Setup

The propagating durations of the four lines are sharply characterized to create a synchronization pattern applied to the switches and measuring devices. By changing synchronization pattern, we change the stringing of the spans and nodes; hence, we can simulate any connection.

The flatness of the output spectrum is used as an indicator to independently set the powers in each fiber line (launched powers in the line fiber and in the dispersion-compensating fiber). The flatness changes because of the variation in the power at the input of the EDFA. The powers are tuned for each line in order to maximize the flatness after a high number of amplifier spans (typically more than 20). Once this tuning is realized, we proceed with the per-band equalization in the node in order to improve the flatness. This per-band equalization is static and is, thus, the same for every connection in the network; it is adjusted after 27 spans of SMF. The signals pass through the nodes after each nine spans, corresponding to the average number of SMF spans between any two nodes of the network. Similarly, we slightly modify this equalization by considering the signal after 18 spans of TeraLight, the signal going through the node after each six spans, corresponding to the average number of TeraLight spans between nodes. The result compromises between

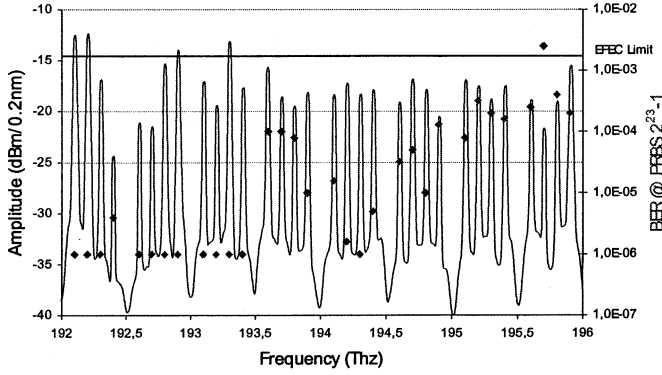


Fig. 5. Focus on Paris→Budapest connection: spectrum (solid line, scale on the left) and BERs (dots, scale on the right) of channels at egress.

the optimal setting of each line. As the matrix switches inside the node are of no use, they were not activated.

We can then see that the multiloop is a simulation tool representing a deployed network though, in a real one, even better BERs could be obtained since they would not suffer from experimental impairments.

- Loop impact: statistical emphasis of polarization mode dispersion and polarization dependent loss and gain-yielding dispersion in BER measurements.
- Multiloop impact: power equalization in the node is the same for every node of every connection, whereas, in a real system, the per-band equalization would be tuned at each node to fit to the incoming signal.

D. Collecting Measurements

Dealing with the BER measurement, the bit sequence used is pseudorandom of length $2^{23}-1$. Only one modulator is used, but the channel spacing—equal to 100 GHz—is sufficiently wide at 10 Gb/s to avoid the use of a second modulator. Because wavebands are taken as the transparent network granularity, we require all the wavelengths of a waveband to show a BER below a given limit. Note that, in practice, the traffic may require only three or less wavelengths between two nodes, but we consider the most constraining condition. We consider two waveband BER limits: $2 \cdot 10^{-3}$, which is the EFEC limit (low margin), and 10^{-6} (high margin).

These features are illustrated in Fig. 5 with the Paris to Budapest connection, in which every channel BER is plotted superimposed to the spectrum. Wavebands 2 and 3 may be used to connect Paris to Budapest with comfortable margins, wavebands 1, 4, 5, 6, and 7 with low margins, while waveband 8 cannot be used because of channel 30.

IV. RESULTS WITH THE FIRST SET OF ENGINEERING RULES

A. Exploitation of These Measures

We have plotted the ratio of wavebands; usable transparently (with low or high margins) depending on the length of the connection (see Fig. 6). For instance, 90% of the wavebands are

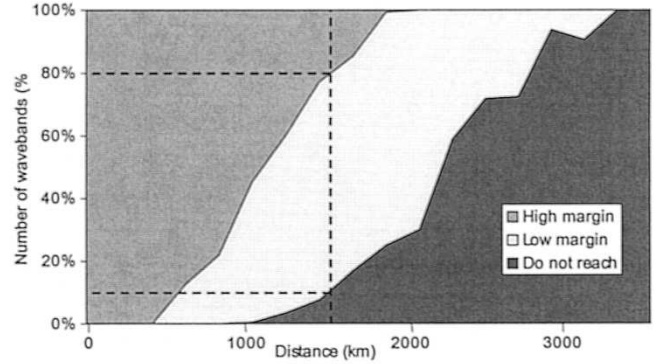


Fig. 6. Ratio of established wavebands versus distance.

able to bridge 1500 km (70% with low margins and 20% with high margins).

For every connection, we consider that the associated traffic can be routed transparently if at least one full waveband can support this connection. Summing the traffic for every valid connection, we thus demonstrate that, with an efficient routing algorithm taking physical constraints into account, up to 99.7% of the total traffic of this network may be routed transparently. If high margins are required, 78.9% may still be routed transparently. The other connections can then use regeneration capabilities of hybrid network.

Of course, the figures given here are the upper limits of what can possibly be achieved with an efficient routing algorithm, at least one taking physical impairments into account [12]. One difficulty is that if a waveband is used to carry the traffic from Madrid to Helsinki (see Fig. 1), the same waveband in the same fiber cannot be used to carry the traffic from Lisbon to Helsinki because these two connections share common fibers. If only one waveband can make it for Madrid to Helsinki, it is likely that only this waveband will be able to make it from Lisbon to Helsinki. That is where optic–electronic–optic (O-E-O) regeneration capabilities can have another interesting role to play. Indeed, the grooming capabilities offered by hybrid cross-connect architectures are interesting features for coping with resources sharing (here, sharing waveband in a succession of fiber links).

Finally, to see the influence of our random drawing of fiber type, we applied the same scheme to the complementary network (i.e., SMF and TeraLight are interchanged), which makes 380 other connections studied. Under these conditions, 99.1% of the traffic may be routed transparently (80.6% with high margins), backing up our former results.

B. Conclusion

This shows that transparency in a 10 Gb/s hybrid heterogeneous pan-European network is feasible using pragmatic physical design: EDFA amplification, no dynamic gain equalization, and constant chromatic dispersion management whatever the waveband and whatever the connection. With the help of an efficient routing algorithm, this system allows up to 99% of the connections established without regeneration.

Actually, the set of chromatic dispersion-management rules used before and their pertinence can be challenged, which is described hereafter.

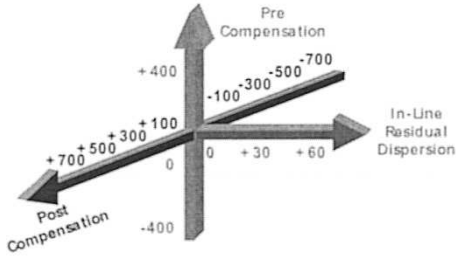


Fig. 7. Dispersion parameter space considered in this experimental work.

V. COMPARISON WITH NUMERICAL SIMULATION WORK

In [5], numerical simulations of a heterogeneous transparent network have been performed with the same simple engineering rules: 100-GHz-spaced 10 Gb/s NRZ signals, EDFAs only, almost the same constant dispersion map whatever the ingress and egress nodes (taking into account realistic variations of this dispersion map), almost the same range of span length (from 60 to 100 km), and the same kind of optical power range (from 0 to 6 dBm launched into the fiber that is close to the power variation considered in the experiments (see Figs. 4 and 5)).

Results have shown that the most favored channel could have a low BER even after experiencing strong nonlinearities, expressed by the nonlinear phase [14]. A nonlinear phase up to approximately 2.5 rad could be obtained, which translates into approximately 3500 km, but most of the channels could only reach approximately 2000 km (1.5 rad nonlinear phase). Hence, the experimental simulation results presented in this paper (see Fig. 6) are consistent with the numerical simulation resultants of [5].

VI. INFLUENCE OF CHROMATIC DISPERSION PARAMETERS

This part of the work studies the impact of dispersion management on the transparently supported traffic in heterogeneous backbone networks. This was achieved by considering different dispersion-management sets of rules and systematically measuring every channel BER of many connections of a network, asking for more than 15 000 new BER measurements [13]. Moreover, we highlight the impacts of network topology and the traffic matrix on the physical design of hybrid networks.

A. Experiment Conditions

A new set of engineering rules, shown in Fig. 7, are considered:

- Precompensation varies from -400 ps/nm to $+400$ ps/nm with a 400 ps/nm step;
- Inline residual dispersion (cumulated dispersion of an amplifier span) is the same for both fiber types and the values taken are $+0$ ps/nm/span, $+30$ ps/nm/span or $+60$ ps/nm/span;
- Postcompensation (inserted dispersion at egress) varies from -700 ps/nm to $+700$ ps/nm with a 200 ps/nm step;
- From now on, only low margins will be considered (BER below 2.10^{-3}).

TABLE I
TRAFFIC VECTORS (PARIS TO OTHER NODES) AND CONNECTION DISTANCES

Node	Lisbon	Madrid	Bern	Rome	Athens	Zagreb	Vienna	Budapest	Prague	Brussels	London
Total fiber length from Paris	1575	1050	525	1275	2400	1425	1125	1800	1275	300	400
Original network traffic contribution	2.7%	10.3%	1.8%	14.7%	2.8%	1.1%	2.0%	2.5%	2.5%	2.5%	15.1%
Extended network traffic contribution	1.8%	6.9%	1.2%	9.9%	1.9%	0.8%	1.4%	1.7%	1.7%	1.7%	10.1%

Node	Dublin	The Hague	Berlin	Vareau	Copenhagen	Stockholm	Oulu	Helsinki	Kiev	Moscow
Total fiber length from Paris	925	500	975	1500	1200	1800	1725	2250	2175	2975
Original network traffic contribution	1.0%	4.1%	20.8%	9.8%	1.4%	2.3%	1.1%	1.3%		
Extended network traffic contribution	0.7%	2.7%	14.0%	6.6%	0.9%	1.5%	0.8%	0.9%	8.4%	24.6%

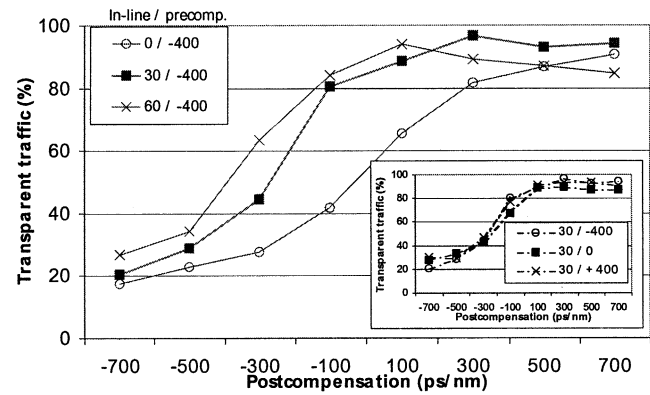


Fig. 8. Proportion of transparently supported traffic versus postcompensation for a precompensation of -400 ps/nm and all inline residual dispersion. The inset shows the influence of the precompensation when the inline residual dispersion is optimized to $+30$ ps/nm/span.

The combination of these data defines the $8 \times 3 \times 3 = 72$ different dispersion-management sets of rules we study in this paper. With the multiloop, we simulated all the connections from Paris to every other city of the network. (Due to time constraints, only these connections were considered.) For each set of dispersion-management rules, the BER was measured for all 19 of the connections and all 32 of the channels, meaning 43 776 measurements.

Table I describes the set of traffic vectors, which actually is a subset of the whole network matrix traffic presented in Section II-B. To evaluate the impact of network topology and the traffic matrix, an extended network was also considered, in which two highly populated and distant nodes (Kiev and Moscow) were added to the original. The original network is then described with its 19 connections, whereas the extended one is described with 21 connections. It can be noted that the traffic between Paris and Kiev plus Paris and Moscow represents 33% of the total. Hence, a strong weight on long-distance connections has been applied in the extended network.

B. Results on the Original Network

First, we consider the original network. The curves in Fig. 8 show this proportion for 24 considered dispersion-management rules: the precompensation is set to -400 ps/nm. In this original

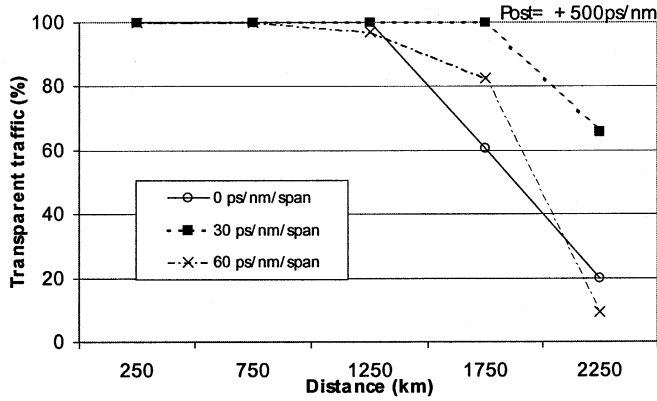


Fig. 9. Proportion of transparently supported traffic versus the distance for the dispersion-management rules with a close-to-optimum postcompensation of +500 ps/nm and precompensation of -400 ps/nm.

network, considering low margins, more than 97% of the traffic may be routed transparently with a proper choice of dispersion management. This is compliant with the results presented in Section IV-A; actually, with the subset of connections Paris to every other nodes. The optimum inline residual dispersion and postcompensation are approximately +30 ps/nm/span and +300 ps/nm, respectively. The inset in Fig. 8 shows the influence of the precompensation from -400 ps/nm to +400 ps/nm when the inline residual dispersion is optimized. Its influence is negligible as compared with the postcompensation. As a matter of fact, the inline residual dispersion is low; hence, the shorter connections of the network corresponding to a linear propagation are not influenced since the cumulated dispersion remains low. For the longest connections, the propagation is nonlinear and postcompensation is known to be critical.

Optimization of the precompensation remains possible, of course, but is not as critical as the postcompensation or the inline residual dispersion over the ranges considered here. Pre- and postcompensation can easily be set with better than 10% accuracy. However, the sources of inline residual dispersion uncertainty are numerous due to the difference for each span between the line dispersion and the compensating fiber dispersion: the value of the line dispersion is not known accurately (because it is often not measured), the dispersion is not perfectly compensated, the slope of the dispersion is not perfectly compensated, the temperature may change the dispersion, etc. Hence, although the optimum postcompensation is equal to +300 ps/nm, the engineering rule on the postcompensation should be preferably set to +500 ps/nm to be more tolerant of inline residual dispersion. As a matter of fact, the proportion of transparent traffic is almost constant for +500 ps/nm postcompensation when the inline compensation varies, which is not the case for +300 ps/nm, since it drops from 97% with +30 ps/nm/span inline residual dispersion to 21% with 0 ps/nm/span (see Fig. 10).

A deeper analysis indicates that the longer the distance that needs to be reached, the sharper the dispersion parameters must be. Fig. 9 shows the proportion of transparent traffic versus distance when maintaining the pre- and postcompensation, respectively, equal to -400 ps/nm and +500 ps/nm, close to their optimum values whatever the inline residual dispersion. Part of the traffic can still be routed transparently after 2000 km

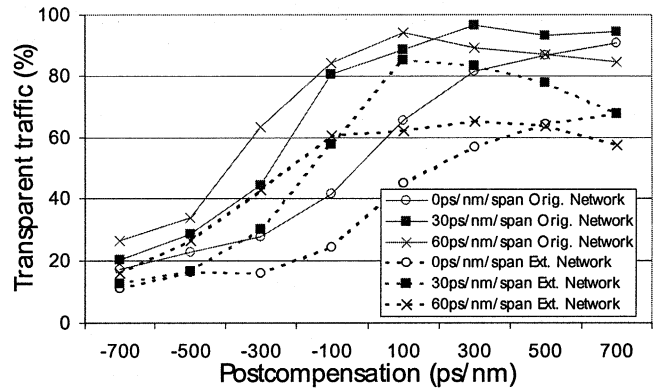


Fig. 10. Variation of the maximum transparent traffic versus postcompensation for different dispersion-management rules and both networks. Precompensation is set to -400 ps/nm. The first three curves are the same as presented in Fig. 8 for the original network and the last three correspond to the extend network.

with an inline residual dispersion of +30 ps/nm/span, while it is impossible for lower or larger values. Hence, the value of inline residual dispersion becomes critical. This was not the case for the aggregated results in Fig. 8 because distances larger than 2000 km only represent a little traffic. As a matter of fact, whatever the inline residual dispersion, more than 87% of the traffic can be routed transparently with +500 ps/nm postcompensation.

C. Network-Design Considerations

The data collected and the traffic matrix defined above give us an analyzing indicator that is the proportion of the traffic that can be routed transparently in this network end-to-end. This indicator quantifies how well the dispersion parameters and the network are suited. When this transparently routed traffic proportion reaches 100%, there is no need for O-E-O regeneration. Nevertheless, it does not directly indicates the number of O-E-O regenerators saved by transparency. There is still a need for a dimensioning working out of these data in order to quantify the savings. Actually, if a long connection cannot be established transparently, at least one regenerator is required along the path, but surely not at each node it passes through.

Dealing with the relative price of network design, it is worth keeping in mind that sharpening dispersion management of a network is theoretically possible. However, the uncertainty with which dispersion parameters are known in deployed networks implies a complete characterization of network links before applying strict dispersion rules. Of course this is not costless and, when urging for a fully transparent network, is likely to cost more than cleverly implementing regeneration capabilities.

D. Rules Applied to the Extended Network

Finally, to evaluate the impact of network topology and traffic matrix, an extended network was also considered where two highly populated and distant nodes (Kiev and Moscow) were added to the original (Table I). We remind readers that a strong weight on long-distance connections has been applied.

The results presented in Fig. 10 show how the pertinence of a set of dispersion-management rules depends on the network

itself (traffic plus topology) as the efficiency of the rules that were compliant with the original network is strongly decreased when considering the extended network. Of course, the physical design that is valid for a large network remains valid for a smaller network, but will surely be overdimensioned and, hence, not cost effective. The choice of the dispersion management becomes more important for the extended network. Compliant with Fig. 9 for a postcompensation of +500 ps/nm, the proportion of transparent traffic "only" decreased from 93% to 78% with a +30 ps/nm/span inline residual dispersion, while it drops from 87% to 64% with either 0 or +60 ps/nm/span.

VII. CONCLUSION

We have experimentally shown that the physical design of a transparent network should be tailored to both the network topology and the network traffic matrix. Hence, both the standard point-to-point transmission and networking aspects should be considered together. For a pan-European network for which 85% of the traffic was shorter than 2000 km, almost all of the connections may be transparent with very simple engineering rules (EDFAs only, no power equalization, except for the per-band equalization in the node, and constant static dispersion management whatever the ingress and egress nodes and whatever the wavelength...). This result is compliant with previous numerical simulation work. We have also shown experimentally how the dispersion management should take unavoidable inaccuracies that are to be found in the field into account.

REFERENCES

- [1] I. Tomkos *et al.*, "Ultra-long-haul DWDM network with 320×320 wavelength-port broadcast & select OXCs," in *Proc. ECOC'02*, Copenhagen, Denmark, Sept. 2002, PD2.1.
- [2] D. Penninckx, G. Charlet, J.-C. Antona, and L. Noirie, "Experimental validation of a transparent waveband-based optical backbone network," in *Proc. ECOC'02*, Copenhagen, Denmark, Sept. 2002, 6.4.4.
- [3] I. Tomkos *et al.*, "Dispersion map design for 10 Gb/s ultra-long haul DWDM transparent optical networks," in *Proc. OECC'02*, Yokohama, Japan, July 2002, PD1.
- [4] A. Ehrhardt, N. Hanik, A. Gladisch, and F. Rumpf, "Field demonstration of a transparent optical 10 Gbit/s-WDM-network based on normalized transmission sections," in *Proc. OFC'02*, Anaheim, CA, Mar. 2002, TuH2, pp. 42–43.
- [5] D. Penninckx and C. Perret, "New physical analysis of 10 Gbit/s transparent optical networks," *IEEE Photon. Technol. Lett.*, vol. 15, pp. 778–780, May 2003.
- [6] R. Pratt, B. Charbonnier, P. Harper, D. Nasset, B. K. Nayar, and N. J. Doran, "40 $\times$ 10.7 Gbit/s DWDM transmission over a meshed ULH network with dynamically re-configurable optical cross connects," in *Proc. OFC'03*, Atlanta, GA, Mar. 2003, PD9.
- [7] P. Pélouso, M. Prunaire, L. Noirie, and D. Penninckx, "Applying optical Transparency to a pan-European hybrid heterogeneous network," in *Proc. OFC'03*, Atlanta, GA, Mar. 2003, PD10.
- [8] L. Noirie, C. Blaizot, and E. Dotaro, "Multi-granularity optical cross-connect," in *Proc. ECOC'00*, vol. 3, Munich, Germany, Sept. 2000, 9.2.4, pp. 269–270.
- [9] J.-P. Faure, L. Noirie, A. Bisson, V. Sabouret, G. Leveau, M. Vigoureux, and E. Dotaro, "A scalable transparent waveband-based optical metropolitan network," in *Proc. ECOC'01*, vol. 6, Amsterdam, The Netherlands, Oct. 2001, P.D. A.1.10, pp. 64–65.
- [10] N. S. Bergano, J. Aspell, C. R. Davidson, P. R. Trischitta, B. M. Nyman, and F. W. Kerfoot, "Bit error rate measurements of 14 000 km 5 Gbit/s fiber-amplifier transmission system using circulating loop," *Electron. Lett.*, vol. 27, pp. 1889–1890, Oct. 10, 1991.
- [11] E. Desurvire and J. Chesnoy, "Alternatives to soliton transmission," in *Proc. Symp. ROSC Physics and Applications of Optical Solitons in Fibers '95*, Kyoto, Japan, Nov. 1995.
- [12] M. Ali, D. Elie-dit-Cosaque, and L. Tancevski, "Network optimization with transmission impairments-based routing," in *Proc. ECOC'01*, vol. 1, Amsterdam, The Netherlands, Sept. 2001, pp. 42–43.
- [13] P. Pélouso, D. Penninckx, and M. Prunaire, "Dispersion management of a pan-European hybrid heterogeneous network," in *Proc. Photonics in Switching*, Versailles, France, Sept. 2003, PS.Tu.B4.
- [14] Y. Frignac and S. Bigo, "Numerical optimization of residual dispersion in dispersion-managed systems at 40 Gbit/s," in *Proc. OFC'00*, vol. 1, Baltimore, MD, Feb. 2000, TuD3, pp. 48–50.



Pierre Pélouso was born in Saint-Etienne, France, in 1976. He received the Engineering Degree from the Ecole Nationale Supérieure de Physique de Marseille, Marseille, France, in 1999.

He started working on optoelectronic devices packaging within Opto+ (Alcatel and France Telecom joint venture), Marcoussis, France, from 2000 to 2001 and then joined the optical networking group of Alcatel Research and Innovation, Marcoussis, France. His current research interests are hybrid backbone networks. He is the author of

seven publications, among them two post-deadline papers in international conferences.



Denis Penninckx was born in Paris, France, in 1970. He graduated from the Ecole Normale Supérieure de Lyon, Lyon, France, in 1993, and received the Engineering Degree from the École Supérieure d'Électricité, Gif/Yvette, France, in 1993 and the Ph.D. degree on optical modulation formats from the École Nationale Supérieure des Télécommunications, Paris, France, in 1997.

He joined Alcatel Research Center, Marcoussis, France, in 1994 to work first on modulation formats and polarization-mode dispersion. He then joined the research team on optical networking, working first on packet switching and now on transparent backbone networks. He has authored or coauthored more than 70 technical papers and 30 patents on all aforementioned topics.

Dr. Penninckx is a Member of the Alcatel Technical Academy and has been in the technical committee of CLEO US from 2001 to 2003.

Magali Prunaire was born in Colombes, France, in 1978. She received the optoelectronic technician diploma from Angers Institut Universitaire de Technologie, Angers, France, in 1999.

She started working on optoelectronic devices packaging within Opto+, Marcoussis, France, from 2000 to 2001 and then joined the optical networking group of Alcatel Research and Innovation, Marcoussis, France. Her current research interests are metro and backbone network.



Ludovic Noirie was born in Montpellier, France, in 1972. He received the engineer diplomas from Ecole Polytechnique, Palaiseau, France, in 1995 and from Ecole Nationale Supérieure des Télécommunications, Paris, France, in 1997.

He joined the Alcatel Research Center, Marcoussis, France, in July 1997 to work on DWDM networks, particularly on optical node design (OXCs and OADMs) and on optical technologies. He has worked on concept for wavebands and optical multigranularity networks. He has authored 29 publications, among them two invited papers on transparent multigranularity optical networks in international conferences.

Mr. Noirie is a Member of the Alcatel Technical Academy.